

Invited: Enhancing Test Quality by Targeting Timing Marginalities Due to Process Variations

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Abstract—IC test methodologies all generate scan tests based on logical stuck-at and timing fault models that assume only a single passive physical defect localized at some circuit node. However, transistors fabricated in advanced technologies are subject to increasing random process variations that can significantly impact multiple devices, and result in highly nonlinear circuit behavior under specific circuit operating conditions. These changes can cause failures missed by current test methods that mostly target passive defects. We make the case for “variation aware” testing of high-performance digital circuits that prioritizes targeting changes in the most performance sensitive device parameters. Our specific focus in this paper is on developing reliable tests for timing failures caused by random variations in transistor threshold voltages.

Keywords—Timing Failures, Silent Error Corruption, Path Delay Testing, Circuit Simulation

I. INTRODUCTION

Traditional stuck-at fault (SAF) and transition delay fault (TDF) models have long been used to generate the input test patterns applied during post manufacturing tests of digital ICs. Since the goal of these tests is to detect and screen out malfunction due to manufacturing defects such as shorts and opens in the circuits, all such logical fault models assume a localized impact of the physical defect, limited to a single circuit node for SAF and TDF tests, or a standard cell in case of the newer Cell Aware Tests (CAT) [1]. This assumption is realistic since in advanced IC processes, manufacturing flaws are relatively rare and physically quite small. Importantly, this assumption of only a single localized defect in any faulty circuit dramatically reduces the exponential number of test patterns that would be otherwise required to test it completely. Fault model-based scan structural testing of even the largest ICs generally takes less than a couple of minutes.

Unfortunately, this single localized fault assumption no longer holds for advanced processing nodes that are experiencing increasing levels of random manufacturing variations due to feature sizes that are only a few atomic dimensions. Note that there are less than 5 silicon atoms in a nanometer length of silicon. With just a handful of atoms and molecules defining each transistor, random fluctuations in their number and orientations in the layout layers that are shaped through photolithography, chemistry and plasma etching can noticeably impact the performance of the individual devices. This can give rise to multiple (even many) “faulty” transistors in the same IC with significant deviations in their electrical parameters from the nominal. Furthermore, these “malfunctioning” transistors do not behave like passive resistive shorts and opens assumed by traditional fault models. Instead,

due to their highly nonlinear operating characteristics, transistor parametric shifts can result in performance failures that are only observed under very specific operating conditions of voltage, temperature and/or frequency. For example, random threshold voltage (V_{th}) variations, that capture the variability in drive strength of individual transistors and are the focus of this paper, can cause identical transistors in a chip, or multiple copies of the same chip, to exhibit very different signal propagation delays, particularly in low voltage operation, resulting in significant delay variations for identical circuit signal paths. Testing for circuit timing compliance in this random variability scenario requires all the circuit path delays to be individually tested, since any path can potentially fail due to an accumulation of delay increases in its transistors. Because of the extremely large number of circuit paths, high coverage scan-based path delay testing is not practical. The practical scan TDF (and CA) timing tests currently employed only target a single localized “lumped” delay defect at individual circuit nodes. These are not sufficient to detect timing failures from the distributed delay effects due to process variations. This limitation is increasingly resulting in test escapes and subsequent failures in operation and is a possible contributor to silent data corruption [2, 3].

While the transistor threshold voltage (V_{th}) is not the only transistor parameter with the potential to adversely impact circuit reliability due to random process variations, it appears to be the most critical and is therefore the focus of this first study. Other transistor parameters, such as drain induced barrier lowering (DIBL), also need to be investigated. Extreme DIBL values can dramatically increase leakage current at high supply voltages with the potential to cause operational errors. Furthermore, the statistical distribution of the DIBL coefficient is known to be log normal, with many more extreme outliers in the tail of the distribution than the normal V_{th} distribution. Clearly, test methods that can screen out failures resulting from parametric shifts due to random manufacturing variations are essential for low nanometer technologies. These parameters must be prioritized based on their likelihood of causing failures and appropriate tests devised to screen out sufficient failing parts to achieve desired defect levels.

In the typical post-manufacturing test flow shown in Fig. 1, structural testing based on scan design-for-test (DFT) techniques is performed first, both before and after the silicon die is packaged. These tests use the scan DFT capability to apply care-



Fig. 1. Manufacturing Test Flow

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-fully crafted test inputs and observe the resulting circuit signals at internal circuit nodes to detect the specific types of faults targeted by the tests. The test content is generated by ATPG algorithms based on static and delay fault models that assume the existence of a single logical fault at some circuit node, as explained earlier. Functional tests are then applied as a final screen to exercise the IC in normal operation over a range of diverse applications and operating conditions to more generally validate correct operation without explicitly targeting specific fault types. While these functional tests can potentially detect faults missed by the structural tests, unfocused functional testing has somewhat limited fault coverage even with long test times. Therefore, any new failure mechanism that is known to be a growing reliability concern must be explicitly targeted by high coverage scan DFT tests developed using effective test generation algorithms. These can then be also followed by functional tests run under operating conditions designed to maximize the likelihood of failure detection. There is growing evidence that random variations in transistor threshold voltage due to manufacturing variability are a significant source of timing failures not targeted by current defect-oriented test methods—an issue that forms the central focus of this paper.

II. BACKGROUND AND RELATED WORK

Table I is extracted from Intel test data [4] published for processors fabricated in 14nm FinFET technology. The data shows fallout for single pattern (static) and two pattern (delay) Cell Aware Tests [1], after the parts were already screened using more traditional SAF and TDF tests. Notice that more than 90% of the additional fallout was from two-pattern CAT delay tests. The majority of these appear to be hard failures that failed at all supply voltages, from the highest VDD down to Vmin, the minimum voltage specified for correct operation at the tested frequency. However, a significant number of the two-pattern failing parts only failed close to Vmin, on average within 55mV of Vmin, while operating correctly at all higher voltages. Because CMOS gates slow down as power supply is lowered, this suggests that these are marginal timing failures, which pass at higher supply voltages. Note that the number of such failures at 1400 DPPM is quite significant, and comprises nearly a third of all the CAT detected failures. Furthermore, since CAT test generation does not target path delay tests [5], some, perhaps even many, timing failures from random process variations may have remained undetected.

It is helpful to consider how likely it is for ICs to contain transistors with extreme values of key device parameters due to process variations. Consider transistor threshold voltage (V_{th}) that critically influences gate delays and is observed to display a normal statistical distribution due to random variations. The probability of any transistor being six standard deviations (6σ) or slower than nominal, based on a shift in V_{th} , is roughly one in a billion, and for a 7σ transistor, about one in a trillion, as shown in Table II. A modern large IC, with billions of transistors, is therefore statistically likely to contain multiple 6σ transistors and occasionally even a 7σ transistor. In fact, failure analysis of production parts has revealed individual transistors with V_{th} shifts as high as 200mV. In combination with other slow transistors, these transistors can cause large increases in circuit path delays. A single extremely slow statistical outlier

TABLE I. ADDITIONAL FALL-OUT FROM CELL AWARE TESTS [4]

CAT-Static Fails (Before Delay Tests)	CAT-Delay Fails (Vmax and Vmin)	CAT-Delay Fails (Vmin only)
400 DPPM	2500 DPPM	1400 DPPM

path from the long tail of the path delay distribution, with mostly slow active transistors can result in a faulty IC. Providing a sufficient guard band in the clock timing margin to protect against all such failures can be cost prohibitive in terms of performance loss. Circuit designers implicitly rely on postproduction testing to screen out such failures. However, as discussed earlier, random gate delay variations can arbitrarily affect any path, and the sheer number of paths in large circuits makes comprehensive timing verification through testing impractical.

Recognizing that high coverage path delay testing is impractical, our approach focuses on identifying distinguishing characteristics of failing paths under specific circuit and operating conditions. As demonstrated in the following sections, under favorable conditions, it is possible to detect timing failures without exhaustively testing every path. This targeted strategy can help lower overall DPPM levels. A key insight driving this approach is that any failing path is an extreme statistical outlier, with a probability of occurrence at least around one in a trillion. If this were not the case, the yield of large SOCs would be essentially zero. To isolate and characterize such extreme paths, we are running large scale circuit simulation experiments over a range of operating voltages—those commonly employed during dynamic voltage frequency scaling for thermal management in high performance processors. While simulating trillions of circuit paths in SPICE is impractical, we expect to generate sufficient data to meaningfully extrapolate the results and evaluate our innovative test approach. Meanwhile, the rest of this paper explains our methodology for detecting timing failures caused by process variations using analytical modeling.

III. V_{th} VARIATION AND GATE DELAYS

In advanced technologies, in addition to manufacturing defects, timing failures in digital circuits can occur due to changes in gate delays due to manufacturing-induced V_{th} variations. This effect becomes especially pronounced as the supply voltage approaches or drops below the threshold voltage, in the near threshold region. Empirical threshold voltage measurements across large populations of modern FinFET transistors have shown a normal distribution extending to least ± 5 standard deviations from nominal [6]. As a result, while individual transistor delays vary, most long circuit paths—with many transistors—still exhibit near-nominal delays due to statistical averaging of these random variations. Nevertheless, extreme outlier slow paths remain statistically likely in large ICs, including those containing single exceptionally slow transistors exhibiting threshold voltage shifts of 7σ or greater. Older technologies with lower variability allowed such outliers to be effectively managed through conservative clock frequencies (timing margins), but low-nanometer technologies face unacceptable performance losses with this approach, necessitating more aggressive clock frequencies. Unfortunately, as previously discussed, existing scan test methodologies and at-speed functional system-level tests cannot reliably detect all timing failures resulting from these random and distributed transistor delays.

Table II, reproduced from [7], summarizes the impact of threshold voltage (V_{th}) variations on transistor switching delays relative to a nominal transistor at two representative supply voltages: 1000mV and 600mV. The nominal threshold voltage (V_{th0}) is 400mV, and manufacturing-induced V_{th} variations are modeled as normally distributed with a typical standard deviation (σ) of 25mV. Delay impacts were computed using a simplified Sakurai-Newton alpha-power law model [8], which approximates switching delay as proportional to $1/(VDD-V_{th})^\alpha$. For ease of calculation and conceptual clarity, α is set to 1, simplifying the delay relationship to $1/(VDD-V_{th})$. This simplification remains reasonably accurate for our illustrative purposes. Similarly, the low probability numbers associated with high V_{th} deviations in Table II have also been rounded for simplicity.

Table II shows two important observations essential to our analysis. First, the increase in transistor delay resulting from a rise in V_{th} is larger than the corresponding speedup from an equivalent decrease in V_{th} . Consequently, even in long paths, the changes do not average out even though the V_{th} distribution is symmetric. Path delays tend to increase as variability grows. Second, gate delays show significantly greater variability at lower supply voltages, especially as VDD approaches the transistor threshold voltage. Note, however, that the accuracy of this simplified Sakurai-Newton model is greatly reduced as VDD approaches V_{th} .

Analyzing delay increases and associated probabilities from Table II, it was conjectured in [7] that at very low voltages the slowest paths predominantly contain a single transistor with an extremely high threshold voltage. Intuitively, this is because a substantial delay increase (such as greater than 600%) is statistically much more likely to originate from one exceptionally slow transistor (such as 7σ , probability approximately one in a trillion, from Table II) rather than multiple moderately slower transistors (such as two 6σ transistors, each with probability around one in a billion), each contributing a smaller delay increase.

For low voltages, if this extreme transistor conjecture can be validated, paths that fail timing due to process variations can be expected to exhibit lumped-delay due to the single extreme transistor and can therefore be tested by conventional TDF and CAT delay tests. The analysis in this paper is therefore aimed at understanding the behavior of statistically extreme slow circuit paths due to process variations over a range of realistic operating voltages. Of particular interest is the inflection point in the power supply, below which timing failure appears to result from lumped delay caused by a single extreme transistor, and above which it is dominated by distributed delays from multiple moderately slow transistors.

TABLE II. GATE DELAY VS. ΔV_{th} AND OCCURRENCE PROBABILITIES

ΔV_{th}		Delay Change (%) due to $\pm \Delta V_{th}$				Probability of $V_{th} > V_{th0} + \Delta V_{th}$
σ	mV	$VDD=1000mV$		$VDD=600mV$		
0	0	0	0	0	0	1 in 2
1	25	-4.0	+4.3	-11.1	+14.3	1 in 6
2	50	-7.7	+9.1	-20.0	+33.3	1 in 44
3	75	-11.1	+14.3	-27.3	+60.0	1 in 740
4	100	-14.3	+20.0	-33.3	+100.0	1 in 32,000
5	125	-17.2	+26.4	-38.5	+166.7	1 in 3.5 Million
6	150	-20.0	+33.3	-42.9	+300.0	1 in 1 Billion
7	175	-22.7	+40.9	-46.7	+700.0	1 in 1 Trillion

IV. IMPACT OF V_{th} VARIATION ON PATH DELAYS

In this section we use Table II, which shows the effect of V_{th} variations on individual transistor (and therefore gate) delays, to study path delays. We are particularly interested in comparing paths containing a single high- σ transistor exhibiting lumped delay behavior with those containing several moderate- σ transistors and therefore exhibiting distributed delay. In this brief paper, we consider relatively short paths comprising six-gate inverter chains, where every transistor has a different V_{th} due to process variations. We assume ideal circuit behavior for clarity: the PMOS and NMOS devices have identical behavior, gate delays are independent and additive, and the downstream effects of V_{th} changes in transistors are neglected.

Table III shows a set of 12 paths with V_{th} values chosen so as to approximately equalize their delays at VDD=700mV. Each path is labeled with a name (e.g., 'P3') indexed in the first column, and its corresponding σ -values are provided in the second column, for example, the σ -values for the six active transistors in P3 appear in the second column of row 3. In the subsequent columns of Table III, the paths are ordered from slowest to fastest based on total path delay at each of several power supply voltages, ranging from 1000mV to 500mV. The nominal transistor threshold, V_{th0} , as in Table II, is 400mV.

Observe from Table III that the slowest ranked path, No. 1 at the top of the table, changes at different voltages. This is a consequence of the fact that, as can be seen in Table II, the percentage delay increase for a given change in supply voltage is different for each gate depending on the specific σ -value of the driving transistor, i.e., the path delay is not a function of supply voltage alone. Furthermore, Table III also shows that the change in rank for many paths is not even monotonic. P10, for example, first moves down in rank as supply voltage is lowered and then rapidly moves up. This observation has significant test implications for circuits that are operated at multiple supply voltages to manage power through dynamic voltage frequency scaling (DVFS). In the face of significant process variations, such parts may need to be retested at each operating frequency since the slowest path can be different at different voltages.

A second important insight from Table III comes from studying P12, which has mostly near-nominal transistors except for a single high $\sigma=5$ transistor. From 1000mV down to 725mV, P12 ranks last among the slowest paths—i.e., it is the fastest path. But then it very quickly improves its rank with decreasing voltage, becoming the slowest path for VDD=675mV—over a span of just 50mV. This behavior highlights how the lumped

TABLE III. CHANGE IN PATH DELAY RANK NO. WITH SUPPLY VOLTAGE

No.	Vth Shifts (σ) for Transistors in Path						1000 mV	800 mV	725 mV	700 mV	675 mV	500 mV
1	1.8	1.7	1.3	1.3	1.2	0.6	P1	P2	P2	P6	P12	P12
2	2.0	1.5	1.4	1.2	1.0	0.8	P2	P1	P1	P4	P11	P11
3	1.9	1.8	1.4	1.4	1.2	0.1	P3	P4	P4	P11	P10	P10
4	1.9	1.8	1.6	1.5	0.6	0.4	P4	P3	P6	P2	P6	P9
5	2.0	1.8	1.5	1.2	1.1	0.2	P5	P6	P9	P9	P9	P8
6	2.0	1.9	1.4	1.3	0.9	0.3	P6	P5	P8	P8	P8	P6
7	1.9	1.8	1.8	1.2	0.6	0.5	P7	P8	P5	P12	P4	P7
8	1.9	1.9	1.8	0.9	0.7	0.6	P8	P9	P3	P10	P7	P5
9	2.0	2.0	1.4	1.0	1.0	0.4	P9	P7	P7	P5	P5	P4
10	1.9	1.9	1.9	1.4	0.5	0.1	P10	P11	P11	P7	P3	P3
11	2.0	2.0	1.7	1.4	0.3	0.3	P11	P10	P10	P3	P2	P2
12	5.0	0.3	0.1	0.0	0.0	0.0	P12	P12	P12	P1	P1	P1

TABLE IV. INFLECTION VOLTAGE-AWARE TEST STRATEGY

Supply Voltage	Test Approach
Above Inflection Voltage	At-speed functional at all Voltages
Near Inflection Voltage	At-speed functional + TDF Tests
Below Inflection Voltage	TDF Tests at Lowest Voltage

delay due to high- σ transistors becomes rapidly more pronounced below an inflection supply voltage, in this case around 675mV. This occurs because the elevated V_{th} of the transistor from process variations (525mV in this instance) approaches VDD in the denominator of the Sakurai-Newton delay equation, greatly increasing the transistor delay. This highly nonlinear increase causes the delay impact of the high- σ transistor to increase rapidly and begin contributing the bulk of the path delay. It thereby causes a lumped delay in the path.

Note that this shift from distributed to lumped delay behavior in the slowest circuit paths with decreasing supply voltage highlights how V_{th} variation not only affects path delays but also dictates the dominant timing failure type at each voltage. In the following section, we leverage these insights to outline a test strategy for variability-induced timing failures across the full DVFS range, although it is particularly efficient for screening out timing failures below the inflection voltage.

V. IMPLICATIONS TO TEST

The dominance of distributed-delay paths at higher voltages, and lumped-delay paths at low voltages, below the inflection point, motivates an adaptive, supply voltage-aware testing approach that efficiently targets the most common failure type at each operating voltage. An important additional factor that our analysis has highlighted is that the slowest path is typically different at each DVFS operating voltage, as seen in Table III. This is due to the nonlinearity of the delay increases for transistors with varying σ values as power supply is lowered. *Prima facie*, this suggests that timing tests must be repeated at each voltage for reliable screening of defective parts.

The analysis presented in this paper offers promising results for efficiently testing low voltage (low power) operating modes, below the inflection voltage. At these voltages, the slowest variability timing paths have been shown to contain a single exceptionally slow transistor with a large V_{th} shift. As a result, the path delay increase is concentrated as a lumped delay at this transistor, and traditional node-oriented scan-based TDF timing tests, including CAT and timing-aware tests [9] remain effective and can be efficiently employed. Importantly, the probability of occurrence of such extreme transistors, estimated to be in the range of $6.5\text{--}8\sigma$ for leading edge circuits, is extremely small, at best one in tens of billions. This ensures that the likelihood of finding more than one such transistor in a single chip is vanishingly small—there can only be one in a failing chip, and therefore a single slow failing path below the inflection voltage. This eliminates the need for testing at multiple voltages. A single high-coverage scan timing test at the lowest voltage, where path delays are most accentuated, is sufficient to screen out such an isolated extreme transistor and ensure failure free operation at all voltages below the inflection point.

In contrast, at higher supply voltages above the inflection voltage, we have shown that delays are typically distributed across multiple transistors with moderate V_{th} variations. The resulting delay behavior is much less localized, greatly reducing

the effectiveness of traditional node-based tests. Furthermore, the many moderate transistors making up these paths are plentiful in a large IC containing billions of transistors. This results in multiple slow paths with similar delays. Different paths can realistically be expected to fail at different supply voltages, particularly since the timing margins used to guard band against process variations and circuit noise are not equal at all voltages. In such a scenario, comprehensive path delay tests are indicated and must be repeated at each operating voltage. In the absence of practical scan path delay tests, at-speed functional testing must be relied upon, despite its high costs. This test strategy is summarized in Table IV. Note that near the inflection point, both scan and functional tests may be needed.

VI. CONCLUSION

Digital circuits in advanced technologies are subject to increasing random process variations that impact all transistors in the design to varying degrees. With billions of circuit paths in large SOCs, timing failures from an accumulation of increased delays in extreme statistical outlier paths are becoming more common. Unfortunately, the path delay tests required to test such failures remain elusive. In this paper we have shown that key properties of these extreme slow paths in a circuit change with supply voltage about an inflection point. At higher voltages, the increased delays are relatively evenly distributed among many transistors in the slowest paths, which necessitates path delay tests. However, below the inflection voltage, nearly all the delay increase is focused on a single transistor and is thus detectable by traditional TDF testing. We outline a strategy that exploits this transition in delay behavior to guide test practical test strategies against timing failures from process variations.

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